

△ The VC-dimension of perceptron $\Rightarrow$

$\Rightarrow$ we want to prove that the VC-dimension of perceptron in $\mathbb{R}^d$ is $d+1$, with 2 steps $\Rightarrow$

① There exist $d+1$ points that perceptron can shatter $\Rightarrow$

Suppose the perceptron $f(x)$:

$$f(x) = \begin{cases} 1 & \text{if } w^T x + b > 0 \\ -1 & \text{otherwise} \end{cases}$$

Consider $d+1$ points $x^{(0)} = (0, \dots, 0)^T$, $x^{(1)} = (1, 0, \dots, 0)^T$,
 $x^{(2)} = (0, 1, 0, \dots, 0)^T$, $\dots$, $x^{(d)} = (0, 0, \dots, 1)^T$

After these $d+1$ points being arbitrarily labeled

$$\Rightarrow y = (y_0, y_1, \dots, y_d)^T \in \{-1, 1\}^{d+1}$$

let $b = 0.5y_0$ and $w = (w_1, w_2, \dots, w_d)$ where $w_i = y_i$,
 $i \in \{1, 2, \dots, d\}$. Thus $f(x)$ can label all these $d+1$ points correctly. So the VC-dimension of perceptron is at

least $d+1 \Rightarrow \underline{\text{VC-dimension}} \geq d+1 \quad \underline{=}$

② Expand $x \in \mathbb{R}^d$ to $X \in \mathbb{R}^{d+1}$, by letting $(X)^T = (x^T, 1)$, and
let $W^T = (w^T, b)$. Thus $\Rightarrow$

$$f(X) = \begin{cases} 1 & \text{if } W^T X > 0 \\ -1 & \text{otherwise} \end{cases}$$

Assume there exist $d+2$ points that perceptron in $\mathbb{R}^d$ can
shatter, namely $x^{(1)}, x^{(2)}, \dots, x^{(d+2)} \in \mathbb{R}^d$ corresponding to
 $X^{(1)}, X^{(2)}, \dots, X^{(d+2)} \in \mathbb{R}^{d+1}$.

Since $d+2$ points in $\mathbb{R}^{d+1}$, there exists certain i such that:

$$X^{(i)} = \sum_{j \neq i} a_j \cdot X^{(j)}$$

where at least one $a_j \neq 0$.

Let $S = \{j | j \neq i, a_j \neq 0\}$. $\forall j \in S$, we give $x^{(j)}$ the label $\text{sign}(a_j)$, and give $x^{(i)}$ a label -1 .

By our assumption, there exists w that make $f(x)$ label those $d+2$ points correctly.

So $\forall j \in S$, we have

$$\Rightarrow a_j \cdot w^T x^{(j)} > 0 \quad \text{and} \quad w^T x^{(i)} \leq 0.$$

also we have $\Rightarrow$

$$\begin{aligned} -w^T x^{(i)} &= w^T \left(\sum_{j \in S} a_j \cdot x^{(j)} \right) \\ &= w^T \left(\sum_{j \in S} a_j \cdot x^{(j)} \right) \\ &= \sum_{j \in S} a_j \cdot w^T x^{(j)} > 0 \end{aligned}$$

So our assumption is False! The VC-dimension of perceptron in $\mathbb{R}^d$ is at most $d+1$

$$\Rightarrow \underline{\text{VC-dimension} \leq d+1} \quad \underline{\underline{2}}$$

With $\underline{1}$ and $\underline{2}$ we can conclude that VC-dimension of perceptron in $\mathbb{R}^d$ is $d+1$.

$$\begin{cases} \text{VC-dimension} > d+1 & \textcircled{1} \\ \text{VC-dimension} \leq d+1 & \textcircled{2} \end{cases} \Rightarrow \text{VC-dimension} = d+1$$